

DECISIONS

DECISIONS.md Architectural decision log for ClauseGuard. One entry per significant decision. Each entry records the date, the decision, alternatives considered, and the rationale. Future decisions are appended. Decisions are not deleted, only superseded. D-001. Project name: ClauseGuard Date. 2026-05-25 Decision. Name the project ClauseGuard. Alternatives. BillScope, ClauseCheck, PolicyLens, GhanaBills, BillsRA. Rationale. Lex is the standard short form for law in academic and legal contexts. Lens is the metaphor that runs through the founder's portfolio (GreenLens for ESG, ClauseGuard for legislation). The "Lens" suffix carries the explainability-first design philosophy across projects. Short, memorable, pronounceable in Ghana and the UK. D-002. Primary user: industry association analyst Date. 2026-05-25 Decision. Build v1 for a mid-level analyst at a Ghanaian industry association (Ghana Chamber of Telecommunications, Fintech Association of Ghana, Ghana ICT and Digital Professionals Association). Alternatives. Parliamentary committee staff, civil society organisations, journalists, individual tech professionals. Rationale. Industry analysts are repeat users with a recurring deadline (position papers within seven to fourteen days of gazette). They care about all eleven risk categories across multiple member types. If the tool works for them, journalists and parliamentary staff adopt it as secondary users. Civil society and individuals are v2. D-003. Two-corpus RAG, Constitution as fixed spine Date. 2026-05-25 Decision. The retrieval architecture is two corpora. The 1992 Constitution is indexed once as the fixed reference. Each bill is the variable input. Every clause is tested against the Constitution. Alternatives. Single corpus with Constitution and bills mixed. Constitution as a second pass after risk classification. Rationale. The Constitution is the ground truth for any Ghanaian risk analysis. Making it the spine forces

every score to be defensible against constitutional text. The alternative (second pass) buried the most important check. Inverting the framing fits the domain. D-004. Clause IDs as primary citations, page numbers as fallback Date. 2026-05-25 Decision. Cite "Clause 12(3)(b)" not "page 8 paragraph 4" . Page numbers stored only as fallback metadata. Alternatives. Page numbers as primary. Mixed citation style. Rationale. Bills from different ministries have different PDF layouts. Page numbers are brittle. Clause IDs are stable and are what lawyers and policy analysts actually cite. This is the standard adopted by Ghanaian parliamentary practice. D-005. Multi-label classification, not single-label Date. 2026-05-25 Decision. One clause can have one or more of the eleven risk categories. Classification is multi-label. Alternatives. Single label per clause. Hierarchical labels. Rationale. Real bills create multiple risks per clause. NITA Bill clause 46 creates jobs risk, constitutional risk, and market risk simultaneously. Forcing a single label loses information the analyst needs. D-006. Severity and likelihood as separate ordinal bands, not a single score Date. 2026-05-25 Decision. Each risk has a severity band (low, moderate, high, severe) and a likelihood band (rare, possible, likely, certain). No single 1 to 5 risk score. Alternatives. Single 1 to 5 score per category. Numeric severity multiplied by numeric likelihood. Rationale. A single number hides disagreements about weighting. Bands are ordinal and resist false precision. Stakeholders can weight the two dimensions themselves through the interface. This protects against bias accusations. D-007. Stakeholder lenses, not a single bill score Date. 2026-05-25 Decision. No default overall bill score. The interface lets the user pick a stakeholder lens (industry, civil society, individual professional, state). The lens applies weights across categories. Alternatives. Single overall score. Multiple visible scores with

no lens selection. Rationale. A single score invites political pushback. Lenses make the weighting visible by construction. An industry user will see a different headline than a civil society user, and the difference is auditable. This is the explainability principle applied to scoring itself. D-008. Eleven risk categories plus two cross-cutting Date. 2026-05-25 Decision. Eleven primary categories: market and competition, jobs and livelihoods, litigation and judicial review, regulatory and compliance, liquidity and capital access, cybersecurity and data breach, software and technical, operational and implementation, constitutional and rights, fiscal risk to state, capture and discretionary power. Two cross-cutting: digital divide, informal sector shock. Alternatives. Borrowed enterprise risk taxonomies (COSO, ISO 31000). Smaller set of five or six categories. Rationale. Off-the-shelf frameworks do not map cleanly onto Ghanaian legislation. The eleven categories were derived from the founder's NITA worked example and stress-tested against three other MoCDT bills. The cross-cutting categories capture Ghana-specific patterns (informal tech workforce, rural and urban divide) that standard frameworks miss. D-009. Human in the loop is the v1 default Date. 2026-05-25 Decision. v1 is semi-automated. The system proposes risks and citations. The analyst reviews, edits, or rejects before export. Alternatives. Full automation. Read-only system with no analyst editing. Rationale. Legal text is high-stakes and the founder is not a lawyer. Full automation would surface plausible-sounding but legally nonsensical outputs. Human in the loop manages this risk while the labelled dataset grows. Move toward more automation only when inter-rater agreement supports it. D-010. Inter-rater agreement gate at Cohen's kappa 0.6 Date. 2026-05-25 Decision. Before scaling labelling

beyond thirty clauses, compute Cohen's kappa between the founder and the consulting lawyer. If below 0.6, stop and sharpen the taxonomy before labelling more. Alternatives. No agreement check. Lower threshold (0.4). Higher threshold (0.8). Rationale. GreenLens stalled because labelling quality issues were not caught early. A 0.6 kappa is the standard threshold for substantial agreement in social science annotation work. Lower lets noise into the dataset. Higher is unrealistic for legal classification.

D-011. Repo organised ministry, year, bill Date. 2026-05-25 Decision. The docs/bills/ tree is structured ministry/year/bill_name/. Each bill folder contains the raw PDF, parsed JSON, metadata, analysis, and evidence sub-folder. Alternatives. Flat structure with metadata tags. Year- first then ministry. Bill-name-first with no hierarchy. Rationale. Audiences ask ministry-first questions. Parliamentary sessions are annual. Bills are the atomic unit of analysis. Ministry then year then bill matches how users think and lets other ministries plug in without restructuring. D-012. Cross-references flagged, not resolved, in v1 Date. 2026-05-25 Decision. The parser detects references to other Acts (NCA Act section 12, Data Protection Act section 47, etc.) and flags them. It does not retrieve the referenced text. The analyst follows the reference manually. Alternatives. Full cross-reference resolution. No detection. Rationale. Full resolution requires indexing every Ghanaian Act the bills might reference. That is a v2 project. Flagging is enough for v1 because the analyst already does this work and only needs the system to surface the reference. D-013. Streamlit for v1 review interface Date. 2026-05-25 Decision. Use Streamlit for the v1 analyst review interface. Migrate to React for v2. Alternatives. React from the start. Pure CLI with markdown output. Rationale. The review interface is the slowest part to build. Streamlit gives a working UI in days, not weeks, and the analyst workflow does not need React-level interaction in v1. The methodology and parsing matter more than the frontend at this stage. This contradicts the founder's GreenLens decision to skip Streamlit. The difference is that ClauseGuard needs an internal review tool first, where GreenLens needs a public demo. D-014. Chroma local, not Pinecone Date. 2026-05-25 Decision. Use Chroma as the vector database, running locally. Alternatives. Pinecone, Weaviate, FAISS, pgvector. Rationale. The corpus is small. Fifteen bills plus the Constitution is under fifty thousand chunks. Chroma runs locally with no cloud cost, no API key, no rate limits. Pinecone is overkill at this scale and adds external dependency. Migrate later if the corpus grows beyond a million chunks.

D-015. NITA Bill as the launch artefact Date. 2026-05-25 Decision. The first public report is on the National Information Technology Authority Bill, 2025. Published before the platform is fully built. Alternatives. Wait for all 15 bills before publishing anything. Pick a less politically charged bill. Rationale. The NITA Bill is already on social media. The information gap between outrage and clause- cited analysis is wide. Shipping a single excellent report on the bill while it is in public debate builds credibility that cannot be manufactured later. The risk is political reaction (logged under D-016).

D-016. Non-partisan stance, documented at the project level Date. 2026-05-25 Decision. ClauseGuard is non-partisan analysis. It cites the bill, cites the Constitution, and lets the evidence make the case. It does not advocate, predict votes, or recommend positions. Alternatives. Explicit advocacy stance. Civil society alignment. Neutral but with named partner organisations. Rationale. Non-partisan is the only stance compatible with being trusted by industry, civil society, parliamentary staff, and the ministry simultaneously. The moment ClauseGuard picks a side, half the audience stops trusting it. This is a hard project-level commitment, not a per-report choice. Template for future decisions ## D-NNN. Short title **Date.** YYYY-MM-DD **Decision.** One sentence. **Alternatives.** What else was considered. **Rationale.** Why the chosen option won.